

# Functional Specification

Version 1.0

## *Order Tracking Systems*

## *Enhancement*

### *OTS & FPP-FKP Integration*

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Date

*10-Feb-2023*

## 1. REVISION HISTORY

| Version | Date        | Author(s)    | Description      |
|---------|-------------|--------------|------------------|
| 1.0     | 10.Feb.2023 | Adhi Rizaldi | Initial document |
|         |             |              |                  |
|         |             |              |                  |

## 2. DOCUMENT APPROVAL

| Full Name                  | Job Title            | Signature                                                                             | Signature Date |
|----------------------------|----------------------|---------------------------------------------------------------------------------------|----------------|
| Ellys Marchiella           | Head of PM           |                                                                                       |                |
| Handrio Purnomo<br>Siregar | RM Supervisor        |  | 22 feb 23      |
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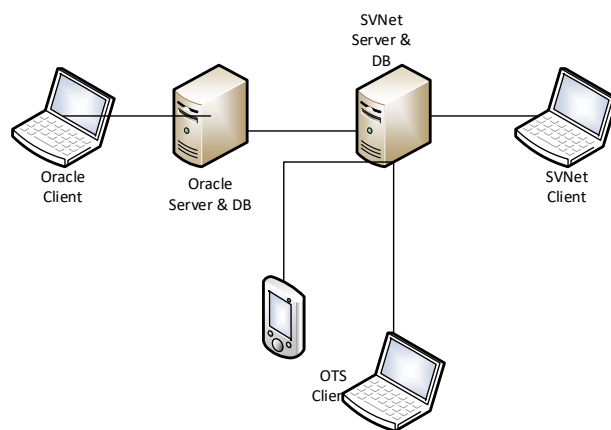
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## 4. OBJECTIVE

Dokumen fungsional specification ini dibuat untuk memberikan informasi yang cukup bagi tim Systems Development/Technical dalam membangun desain teknis untuk setiap fungsi yang ada dalam aplikasi, dan yang akan dikerjakan sebagai bagian dari requirement. Dokumen ini juga sebagai konfirmasi untuk memenuhi user requirement.

## 5. APPLICATION LANDSCAPE

### 5.1 Architecture



## 6. FUNCTIONAL ASPECT

### 6.1 Flow Process

Proses yang nantinya akan terjadi setelah adanya Order Tracking Systems Enhancement ini adalah sebagai berikut:

- Tim Procurement SHP akan me-maintain data item-item yang akan di-track di aplikasi OTS, per supplier, **pada menu ASL Oracle-EBS**.
- Data outstanding PO** yang ada di aplikasi OTS, adalah hanya PO-PO Outstanding, yang itemnya masuk kategori a.) di atas, sehingga Supplier hanya akan perlu mengupdate data-data yang tampil di form OTS tersebut saja.
- Setiap hari, database OTS yang ada di platform SVNNet tersebut, akan di-backup ke dalam tabel history, sehingga data yang tersimpan di database tersebut adalah sepanjang 250 hari kalender, atau **kurang lebih selama 8 bulan**.
- Data **historical**, hanya **dapat di-View**, berdasarkan parameter-parameter variable tertentu. Tidak bisa diedit.
- Jika terjadi perbedaan antara data tanggal **Revised PO Need By Date** yang di-input, dengan data tanggal **Need By Date** yang ada di Oracle-EBS, maka akan terjadi proses pembuatan dokumen FPP-FKP secara otomatis, langsung approved, namun ada identifikasi bahwa dokumen tersebut di-inisiasi dari OTS.
- Otomatis, data tanggal Need By Date di **Oracle-EBS akan terupdate** sesuai dokumen FPP-FKP tersebut.

- g. Secara berkala, sekumpulan data OTS dapat diolah sedemikian rupa sehingga dapat secara otomatis dengan logika perhitungan tertentu mengupdate master data **Processing Leadtime** di Oracle-EBS. Pada akhirnya hasil update tersebut dijadikan inputan pada proses planning di **aplikasi Blue Yonder**.

## 6.2 Application

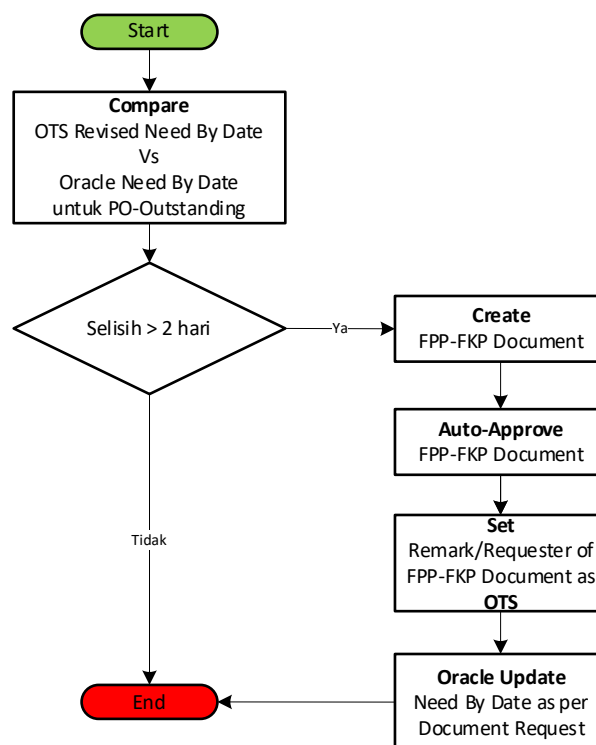
Aplikasi/modul yang akan dibangun/direvisi pada dokumen Functional Design ini adalah di bagian Web-Application, yaitu adanya proses pembuatan dokumen FPPFKP secara otomatis di modul SVNNet pada saat kondisi terjadinya perbedaan informasi tanggal **Need By Date** yang ada di Oracle, dibandingkan dengan tanggal **Revised Need By Date** yang ada di modul OTS, untuk suatu PO-Line tertentu.

Jadi, harapannya terjadi integrasi antara modul OTS dengan modul FPPFKP, sehingga informasi Need By Date di Oracle-EBS untuk dokumen Purchase Order Line terkait menjadi dapat di-update sesuai kondisi aktualnya di lapangan. Tim PPIC dapat mendapat gambaran yang lebih detail akurat terkait proyeksi kedatangan barang, dalam proses planning-nya.

## 6.3 Design

### 6.3.1 Scheduler Auto-Create OTS - FPP-FKP

#### 6.3.1.1 Process Flow



### 6.3.1.2 Field Description

| No | Field Name   | Description                                           | Data Type | Data         |
|----|--------------|-------------------------------------------------------|-----------|--------------|
| 1  | PO_HEADER_ID | ID for Purchase Order Number                          | Number    | As per query |
| 2  | PO_LINE_ID   | ID for Purchase Order Line Number                     | Number    | As per query |
| 3  | NEED_BY_DATE | Date when Goods needs to be arrived at Warehouse Dock | Datetime  | As per query |

### 6.3.1.3 Action Control

| No | Control Name | Description               | Result          |
|----|--------------|---------------------------|-----------------|
| 1  | Scheduler    | To Create FPPFKP Document | Daily Scheduled |
|    |              |                           |                 |

### 6.3.1.4 Query Logic

```

SELECT osp.location_code,
       osp.po_header_id,
       osp.po_number,
       osp.line_num,
       osp.shipment_num,
       osp.status,
       osp.vendor_id,
       osp.gbpa,
       osp.vendor_name,
       osp.vendor_site_code,
       osp.po_line_id,
       osp.item_id,
       osp.item_code,
       osp.item_description,
       osp.vendor_product_num,
       osp.ship_to_organization_id,
       osp.need_by_date,
       NVL (ots.dtm_revised_po_need_bydate, osp.need_by_date)
         dtm_revised_po_need_bydate,
       ( osp.need_by_date
       - NVL (ots.dtm_revised_po_need_bydate, osp.need_by_date))
         gap,
       osp.delivery_date,
       osp.promised_date,
       osp.unit_meas_lookup_code,
       osp.quantity,
       osp.quantity_received,
       osp.po_line_loc_status,
       osp.created_by,
       osp.user_name,
       osp.remarks,
       osp.approved_date,
       osp.action_date,
       osp.description,
       osp.category_id,
       REPLACE (REPLACE (DBMS_LOB.SUBSTR (osp.note, 4000, 1), CHR (13), ' '),
               CHR (10),
               ' '),
       note,
       osp.creation_date,
       CASE WHEN osp.gbpa IS NOT NULL THEN osp.bpa_status ELSE NULL END
         bpa_status,
       osp.end_gbpa
FROM (SELECT hrl.location_code,
            ph.po_header_id,
            ph.segment1 po_number,
            pl.line_num,

```

```

plloc.shipment_num,
INITCAP (ph.authorization_status) status,
ph.vendor_id,
(SELECT phal.segment1
 FROM po_headers_all phal
 WHERE phal.po_header_id(+) = pl.from_header_id)
gbpa,
aps.vendor_name,
assa.vendor_site_code,
pl.po_line_id,
pl.item_id,
(SELECT segment1
 FROM mtl_system_items
 WHERE inventory_item_id = pl.item_id
 AND organization_id = plloc.ship_to_organization_id)
item_code,
pl.item_description,
(SELECT primary_vendor_item
 FROM po_approved_supplier_list
 WHERE item_id = pl.item_id
 AND asl_status_id = 2
 AND vendor_id = ph.vendor_id
 AND ROWNUM < 2)
vendor_product_num,
plloc.ship_to_organization_id,
prla.need_by_date,
plloc.promised_date,
plloc.need_by_date delivery_date,
muom.uom_code unit_meas_lookup_code,
plloc.quantity,
plloc.quantity_received,
plloc.closed_code po_line_loc_status,
ph.created_by,
fu.user_name,
plloc.ATTRIBUTE10 remarks,
(SELECT MIN (ACTION_DATE)
 FROM PO_ACTION_HISTORY_V
 WHERE OBJECT_TYPE_CODE = 'PO'
 AND OBJECT_ID = ph.po_header_id
 AND ACTION_CODE = 'APPROVE')
approved_date,
(SELECT MIN (action_date)
 FROM po_acceptances
 WHERE po_header_id(+) = ph.po_header_id)
action_date,
mc.description,
mc.category_id,
REPLACE (
 REPLACE (DBMS_LOB.SUBSTR (pac.note, 4000, 1),
 CHR (13),
 ' '),
 CHR (10),
 ' ')
note,
ph.creation_date,
NVL ( (SELECT phal.closed_code
 FROM po_headers_all phal
 WHERE phal.po_header_id(+) = pl.from_header_id),
 'OPEN')
bpa_status,
(SELECT phal.end_date
 FROM po_headers_all phal
 WHERE phal.po_header_id(+) = pl.from_header_id)
end_gbpa
FROM po_headers_all ph,
po_lines_all pl,
po_line_locations_all plloc,
ap_suppliers aps,
ap_supplier_sites_all assa,
mtl_units_of_measure_tl muom,
PO_REQUISITION_LINES_ALL PRLA,
po_requisition_headers_all PRHA,
HR_LOCATIONS_ALL_TL HRL,
po_releases_all PRA,
PO_ACCEPTANCES pac,
fnd_user fu,
mtl_categories mc
WHERE ph.org_id = 82
AND ph.po_header_id = pl.po_header_id
AND ph.vendor_id = aps.vendor_id

```

```

AND pl.po_line_id = plloc.po_line_id
AND pac.po_header_id(+) = ph.po_header_id
AND HRL.location_id = plloc.ship_to_location_id
AND ph.vendor_site_id = assa.vendor_site_id
AND muom.unit_of_measure = plloc.unit_meas_lookup_code
AND pra.po_header_id(+) = ph.po_header_id
AND PRLA.LINE_LOCATION_ID(+) = PLLOC.LINE_LOCATION_ID
AND PRHA.REQUISITION_HEADER_ID(+) = PRLA.REQUISITION_HEADER_ID
AND (plloc.quantity - plloc.quantity_received) > 0
AND plloc.quantity_cancelled = 0
AND ph.authorization_status IN ('APPROVED', 'IN PROCESS')
AND NVL (ph.cancel_flag, 'N') <> 'Y'
AND NVL (pl.closed_code, 'OPEN') = 'OPEN'
AND plloc.closed_code NOT IN ('CLOSED FOR RECEIVING', 'CLOSED')
AND ( pra.authorization_status IN ('APPROVED', 'IN PROCESS')
      OR pra.authorization_status IS NULL)
AND ph.created_by = fu.user_id
AND pl.category_id = mc.category_id
AND pac.revision_num(+) = 0
AND pac.action(+) = 'WEB'

UNION
SELECT hrl.location_code,
       ph.po_header_id,
       ph.segment1 po_number,
       pl.line_num,
       plloc.shipment_num,
       INITCAP (ph.authorization_status) status,
       ph.vendor_id,
       (SELECT phal.segment1
        FROM po_headers_all phal
        WHERE phal.po_header_id(+) = pl.from_header_id)
       gbpa,
       aps.vendor_name,
       assa.vendor_site_code,
       pl.po_line_id,
       pl.item_id,
       (SELECT segment1
        FROM mtl_system_items
        WHERE inventory_item_id = pl.item_id
              AND organization_id = plloc.ship_to_organization_id)
       item_code,
       pl.item_description,
       (SELECT primary_vendor_item
        FROM po_approved_supplier_list
        WHERE item_id = pl.item_id
              AND asl_status_id = 2
              AND vendor_id = ph.vendor_id
              AND ROWNUM < 2)
       vendor_product_num,
       plloc.ship_to_organization_id,
       prla.need_by_date,
       plloc.promised_date,
       plloc.need_by_date delivery_date,
       muom.uom_code unit_meas_lookup_code,
       plloc.quantity,
       plloc.quantity_received,
       plloc.closed_code po_line_loc_status,
       ph.created_by,
       fu.user_name,
       plloc.ATTRIBUTE10 remarks,
       (SELECT MIN (ACTION_DATE)
        FROM PO_ACTION_HISTORY_V
        WHERE OBJECT_TYPE_CODE = 'PO'
              AND OBJECT_ID = ph.po_header_id
              AND ACTION_CODE = 'APPROVE')
       approved_date,
       (SELECT MIN (action_date)
        FROM po_acceptances
        WHERE po_header_id(+) = ph.po_header_id)
       action_date,
       mc.description,
       mc.category_id,
       REPLACE (
         REPLACE (DBMS_LOB.SUBSTR (pac.note, 4000, 1),
                               CHR (13),
                               ' '),
         CHR (10),
         ' ')
       note,
       ph.creation_date,

```

```

        NVL ( (SELECT phal.closed_code
                FROM po_headers_all phal
                WHERE phal.po_header_id(+) = pl.from_header_id),
              'OPEN')
        bpa_status,
        (SELECT phal.end_date
         FROM po_headers_all phal
         WHERE phal.po_header_id(+) = pl.from_header_id)
        end gbpa
FROM po_headers_all ph,
     po_lines_all pl,
     po_line_locations_all plloc,
     ap_suppliers aps,
     ap_supplier_sites_all assa,
     mtl_units_of_measure_tl muom,
     PO_REQUISITION_LINES_ALL PRLA,
     po_requisition_headers_all PRHA,
     HR_LOCATIONS_ALL_TL HRL,
     po_releases_all PRA,
     PO_ACCEPTANCES pac,
     fnd_user fu,
     mtl_categories mc
WHERE   ph.org_id = 82
        AND ph.po_header_id = pl.po_header_id
        AND ph.vendor_id = aps.vendor_id
        AND pl.po_line_id = plloc.po_line_id
        AND pac.po_header_id(+) = ph.po_header_id
        AND HRL.location_id = plloc.ship_to_location_id
        AND ph.vendor_site_id = assa.vendor_site_id
        AND muom.unit_of_measure = plloc.unit_meas_lookup_code
        AND pra.po_header_id(+) = ph.po_header_id
        AND PRLA.LINE_LOCATION_ID(+) = PLLOC.LINE_LOCATION_ID
        AND PRHA.REQUISITION_HEADER_ID(+) = PRLA.REQUISITION_HEADER_ID
        AND (plloc.quantity - plloc.quantity_received) > 0
        AND plloc.quantity_cancelled = 0
        AND ph.authorization_status IN ('APPROVED', 'IN PROCESS')
        AND NVL (ph.cancel_flag, 'N') <> 'Y'
        AND NVL (pl.closed_code, 'OPEN') = 'OPEN'
        AND plloc.closed_code NOT IN ('CLOSED FOR RECEIVING', 'CLOSED')
        AND ( pra.authorization_status IN ('APPROVED', 'IN PROCESS')
              OR pra.authorization_status IS NULL)
        AND ph.created_by = fu.user_id
        AND pl.category_id = mc.category_id
        AND pac.action(+) = 'WEB'
        AND pac.revision_num(+) = 0) osp,
(SELECT a.int_order_tracking_id,
       a.int_vendor_id,
       a.int_po_line_id,
       a.txt_po_number,
       a.txt_item_code,
       b.dtm_revised_po_need_bydate
FROM   (SELECT int_order_tracking_id,
              int_vendor_id,
              int_po_line_id,
              txt_po_number,
              txt_item_code
        FROM XXSHP_OTS_T_ORDER_HDR) A,
       (SELECT int_order_tracking_id,
              int_order_tracking_dtl_id,
              txt_item_code,
              dtm_revised_po_need_bydate
        FROM XXSHP_OTS_T_ORDER_DTL) B
WHERE   1 = 1
        AND A.int_order_tracking_id = B.int_order_tracking_id
        AND a.txt_item_code = b.txt_item_code) OTS
WHERE   1 = 1
        AND osp.po_number = ots.txt_po_number
        AND osp.po_line_id = ots.int_po_line_id
        AND osp.item_code = ots.txt_item_code
        AND ( osp.need_by_date
              - NVL (ots.dtm_revised_po_need_bydate, osp.need_by_date)) > 2

```

### 6.3.1.5 Business Rule

- a. Scheduler berjalan tiap hari, pk. 01:00 AM, untuk meng-compare data sesuai query, sehingga jika didapatkan data pada kondisi tersebut, maka akan terbentuk dokumen FPP-FKP dengan status Approved, dengan nama requester adalah OTS, dan dengan perubahan informasi Need By Date sesuai informasi **dtm\_revised\_po\_need\_bydate** dari OTS.

#### 6.3.1.5.1 User Access

| Responsibility | Access |
|----------------|--------|
| As is          | As is  |
|                |        |